

Documentation Positioner DLL ANC350

Version 2.22

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1 General Functions:

1.1 PositionerCheck

Determines number of connected positioners and their respective hardware ids.

Parameter

Return value	[out]	Number of available positioners.
PositionerInfo	[out]	Array of structure PositionerInfo. It contains the hardware id for unlocked devices. Connected devices are marked as locked.

1.2 PositionerConnect

Establishes a connections to the selected device.

Parameter

Return value	[out]	Handle for subsequently accesses to the device.
deviceNo	[in]	Number of the device to connect.

1.3 PositionerSetHardwareId

Sets the hardware id for the device. The hardware id serves as a differentiator in case of multiple simultaneous connected devices.

Parameter

deviceHandle	[in]	Handle of addressed device.
Id	[in]	Hardware Id

1.4 PositionerGetStatus

Determines the status of the selected axis.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Number of addressed Axis (0.. Number of axes - 1)
status	[out]	actual status
Bit 0: (moving) Axis is moving		
Bit 1: (stop detected) Hump detected		
Bit 2: (sensor error) Error of sensor		
Bit 3: (sensor disconnected) Sensor not connected		

1.5 PositionerSetOutput

Activates / Deactivates the output relais of the addressed axis.

Parameter

deviceHandle	[in]	Handle of addressed device.
enable	[in]	Status of relais

1.6 PositionerStaticAmplitude

Sets the output voltage for resistive sensors.

Parameter

deviceHandle	[in]	Handle of addressed device.
amplitude	[in]	Amplitude [mV].

1.7 PositionerDcInEnable

Activates / Deactivates the dc input of the addressed axis. Only applicable for scanner and dither axes.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Status of input

1.8 PositionerAcInEnable

Activates / Deactivates the ac input of the addressed axis. Only applicable for dither axis.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Status of input

1.9 PositionerIntEnable

Activates / Deactivates the internal signal generation of the addressed axis. Only applicable for scanner and dither axes.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Status of internal signal

1.10 PositionerBandwidthLimitEnable

Activates / Deactivates the band width limiter of the addressed axis. Only applicable for scanner axes.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Status of bandwidth limiter

1.11 PositionerGetDcInEnable

Determines the status of dc input of the addressed axis. Only applicable for scanner and dither axes.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[out]	Status of input

1.12 PositionerGetAcInEnable

Determines the status of ac input of the addressed axis. Only applicable for dither axis.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[out]	Status of input

1.13 PositionerGetIntEnable

Determines the status of internal signal generation of the addressed axis. Only applicable for scanner and dither axes.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[out]	Status of internal signal

1.14 PositionerGetBandwidthLimitEnable

Determines the status of the band width limiter of the addressed axis. Only applicable for scanner axes.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[out]	Status of bandwidth limiter

1.15 PositionerCapMeasure

Determines the capacity of the piezo of the addressed axis by measurement.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
capacity	[out]	Measured capacity in [pF]. A value of -1 indicates an error.

1.16 PositionerSensorPowerGroupA

Switches power of sensor group A. Sensor group A contains either the sensors of axis 1..3 or 1..2 dependent on hardware of controller.

Parameter

deviceHandle	[in]	Handle of addressed device.
enable	[in]	on/off

1.17 PositionerSensorPowerGroupB

Switches power of sensor group B. Sensor group B contains either the sensors of axis 4..6 or 3 dependent on hardware of controller.

Parameter

deviceHandle	[in]	Handle of addressed device.
enable	[in]	on/off

2 Positioning functions:

2.1 PositionerGetPosition

Determines the actual position of the addressed axis.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
position	[out]	Position in unit of actor multiplied with 1000

2.2 PositionerGetRotCount

Determines the actual number of rotations in case of a rotary actor.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
count	[out]	Number of rotations

2.3 PositionerGetReference

Determines the distance of reference mark to the origin.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
position	[out]	Reference position in unit of actor
valid	[out]	Indicator for a valid reference position

2.4 PositionerGetReferenceRotCount

Determines the actual number of rotations for the reference position in case of a rotary actor.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
count	[out]	Number of rotations

2.5 PositionerResetPosition

Sets the origin to the actual position

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)

3 Functions for position approach:

3.1 PositionerMoveAbsolute

Starts approach to absolute target position. Previous movement will be stopped

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
position	[in]	Target position in unit of actor multiplied with 1000
rotCount	[in]	Number of rotations in case of rotary actor and deactivated single circle mode

3.2 PositionerUpdateAbsolute

Updates target position for a *running* approach. This function has a lower performance impact on a running approach as PositionerMoveAbsolute.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
position	[in]	Target position in unit of actor multiplied with 1000

3.3 PositionerMoveRelative

Starts approach to relative target position. Previous movement will be stopped.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
distance	[in]	Relative position in unit of actor multiplied with 1000
rotCount	[in]	Number of rotations in case of rotary actor and deactivated single circle mode

3.4 PositionerMoveReference

Starts approach to reference position. Previous movement will be stopped.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)

3.5 PositionerStopApproach

Stops approaching target, relative or reference position of selected axis. DC level of affected axis after stopping depends on setting by *PositionerSetTargetGround*.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)

3.6 PositionerStopDetection

Switches Stop Detection

Parameter

deviceHandle	[in]	Handle of addressed device
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axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Switch (true=on, false=off)

3.7 PositionerSetStopDetectionSticky

When enabled, an active stop detection status remains active until cleared manually by PositionerClearStopDetection.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Switch (true=on, false=off)

3.8 PositionerClearStopDetection

When PositionerSetStopDetectionAuto is disabled, this clears the stop detection status.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)

3.9 PositionerSingleCircleMode

Switches single circle mode. In case of activated single circle mode the number of rotations are ignored and the shortest way to target position is used. Only relevant for rotary actors.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Switch (true=on, false=off)

3.10 PositionerSetTargetGround

When enabled, the actor voltage is set to zero after closed loop positioning finished.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
enable	[in]	Switch (true=on, false=off)

3.11 PositionerSetTargetPos

Sets the target position for the selected axis for use with *PositionerMoveAbsoluteSync*.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
position	[in]	Target position in unit of actor multiplied with 1000
rotCount	[in]	Number of rotations in case of rotary actor and deactivated

single circle mode

3.12 PositionerMoveAbsoluteSync

Starts the synchronous approach to absolute target positions for selected axes. Previous movement will be stopped. The target positions for each axis is defined by *PositionerSetTargetPos*.

Parameter

deviceHandle	[in]	Handle of addressed device
axes	[in]	Bitmask of axes to start

4 Functions for amplitude control

4.1 PositionerAmplitudeControl

Selects type of amplitude control. The amplitude is controlled by the positioner to hold the value constant determined by the selected type of amplitude control.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
mode	[in]	type of amplitude control: 0: Speed, 1: Amplitude, 2: Step Size

4.2 PositionerAmplitude

Sets the amplitude setpoint.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
amplitude	[in]	Amplitude [mV]

4.3 PositionerGetAmplitude

Determines the actual amplitude. In case of standstill of the actor this is the amplitude setpoint. In case of movement the amplitude set by amplitude control is determined.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
amplitude	[out]	Amplitude [mV]

4.4 PositionerGetSpeed

Determines the actual speed. In case of standstill of the actor this is the calculated speed resulting from amplitude setpoint, frequency and motor parameters. In case of movement this is the measured speed.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
speed	[out]	Speed in unit of actor / s * 1000

4.5 PositionerGetStepwidth

Determines the step width. In case of standstill of the motor this is the calculated step width resulting from amplitude setpoint, frequency and motor parameters. In case of movement this is the measured step width.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
stepwidth	[out]	Step width in unit of actor * 1000

5 Functions for manual positioning

5.1 PositionerMoveSingleStep

Starts a one step positioning, Previous movement will be stopped.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
dir	[in]	Direction for positioning (0: forward, 1: backward).

5.2 PositionerMoveContinuous

Starts continuously positioning with set parameters for amplitude and speed and amplitude control respectively.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
dir	[in]	Direction for positioning (0: forward, 1: backward)

5.3 PositionerStopMoving

Stops any positioning. DC level of affected axis is zero after stopping

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)

5.4 PositionerExternalStepFwdInput

Configures external step trigger input for selected axis. A trigger on this input results in a forward single step.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
input	[in]	Trigger input (0: disabled, 1..6 input trigger)

5.5 PositionerExternalStepBkwInput

Configures external step trigger input for selected axis. A trigger on this input results in a backward single step.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
input	[in]	Trigger input (0: disabled, 1..6 input trigger)

5.6 PositionerExternalStepInputEdge

Configures edge sensitivity of external step trigger input for selected axis.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
edge	[in]	(0: raising, 1: falling)

5.7 PositionerStepCount

Configures the number of successive steps caused by external trigger, or manual step request.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
stepCount	[in]	Number of steps (1..65535)

6 Functions for adjustment of positioning parameters

6.1 PositionerFrequency

Sets the frequency.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
Frequency	[in]	Frequency [Hz].

6.2 PositionerGetFrequency

Determines the frequency.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
frequency	[out]	Frequency [Hz]

6.3 PositionerDCLevel

Sets the DC Level.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
level	[in]	DC Level [mV].

6.4 PositionerGetDcLevel

Determines the status actual DC Level.

Parameter

deviceHandle	[in]	Handle of addressed device.
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
level	[out]	DC Level [mV].

7 Functions for trigger configuration

7.1 PositionerTriggerModeIn

Selects the mode of the input trigger signals.

0: Disable. The inputs don't trigger anything.

1: Quadrature. Three pairs of trigger in signals are used to accept AB-Signals for relative positioning.

2: Coarse. The trigger in signals are used to generate coarse steps

Parameter

deviceHandle	[in]	Handle of addressed device
mode	[in]	Mode of trigger pins

7.2 PositionerTriggerModeOut

Selects the mode of the input trigger signals.

0: Disable. The inputs don't trigger anything.

1: Position. The Trigger Outputs reacts to the defined position ranges with the selected polarity

2: Quadrature. Three pairs of trigger out signals are used to signal relative movement as AB-signals.

3: IcHaus. The trigger out signals are used to output the internal position signal of num-sensors

Parameter

deviceHandle	[in]	Handle of addressed device
mode	[in]	Mode of trigger pins

7.3 Trigger Mode Position

7.3.1 PositionerTrigger

Sets the trigger thresholds for the external trigger.

Parameter

deviceHandle	[in]	Handle of addressed device
triggerNo	[in]	Number of addressed Trigger (0.. 5)
lowLevel	[in]	Lower trigger threshold in unit of actor * 1000
highLevel	[in]	Upper trigger threshold in unit of actor * 1000

7.3.2 PositionerTriggerAxis

Selects the corresponding axis for the addressed trigger.

Parameter

deviceHandle	[in]	Handle of addressed device
triggerNo	[in]	Number of addressed Trigger (0.. 5)
axis	[in]	Axis (0..Number of axes – 1)

7.3.3 PositionerTriggerPolarity

Sets the polarity of the external trigger.

Parameter

deviceHandle	[in]	Handle of addressed device
triggerNo	[in]	Number of addressed Trigger (0.. 5)
polarity	[in]	polarity, 0: low active, 1: high active

7.3.4 PositionerTriggerEpsilon

Sets the hysteresis of the external trigger.

Parameter

deviceHandle	[in]	Handle of addressed device
triggerNo	[in]	Number of addressed Trigger (0.. 5)
epsilon	[in]	Hysteresis in unit of actor * 1000

7.4 Trigger Mode Quadrature

7.4.1 PositionerQuadratureAxis

Selects the axis for use with this trigger in/out pair.

Parameter

deviceHandle	[in]	Handle of addressed device
quadratureNo	[in]	Number of addressed quadrature unit (0.. 2)
axis	[in]	Axis (0..Number of axes – 1)

7.4.2 PositionerQuadratureInputPeriod

Selects the stepsize the controller executes when detecting a step on it's input AB-signal.

Parameter

deviceHandle	[in]	Handle of addressed device
quadratureNo	[in]	Number of addressed quadrature unit (0.. 2)
period	[in]	stepsize in unit of actor * 1000

7.4.3 PositionerQuadratureOutputPeriod

Selects the position difference which causes a step on the output AB-signal.

Parameter

deviceHandle	[in]	Handle of addressed device
quadratureNo	[in]	Number of addressed quadrature unit (0.. 2)
period	[in]	period in unit of actor * 1000

8 Functions for actor configuration

8.1 PositionerLoad

Loads a parameter file for actor configuration.

Parameter

deviceHandle	[in]	Handle of addressed device
axisNo	[in]	Handle of addressed axis (0..Number of axes - 1)
file	[in]	File name